Project Title: Predictive Pipeline Analysis: American Airlines Sales & Capacity Optimization

✈️ Project Overview

This project applies data-driven modeling to the aviation industry, simulating the conversion funnel of airline ticket sales and load capacity. By analyzing passenger booking behaviors against seat inventory and seasonal demand, this analysis identifies key drivers in "net yield" performance, demonstrating how analytical modeling can optimize both customer acquisition and flight profitability.

🛠️ Technical Stack

- **Database**: SQLite (Normalized Relational Model)
- **Programming**: Python (Pandas, Seaborn, Matplotlib)
- **Data Strategy**: Synthetic data generation to model booking patterns, loyalty program interactions, and flight capacity metrics.
- **Analysis**: SQL (Complex Joins, Time-Series Grouping, and KPI Calculation)

📊 Key Insights

- **Booking Efficiency**: Analyzed the relationship between loyalty program status (e.g., AAdvantage tiers) and ticket conversion rates.
- **Capacity Optimization**: Visualized the impact of "Load Factor" (percentage of seats filled) on total flight revenue.
- **Demand Forecasting**: Identified high-performing booking channels and booking windows to maximize front-end revenue per passenger.

🚀 Why This Matters

Transitioning from automotive digital retailing to aviation requires a sharp focus on high-velocity data. This project showcases my ability to adapt my domain experience—optimizing lead pipelines and conversion ratios—to the complex, high-stakes environment of airline revenue management and operational data engineering.